

Detailed Project Report (DPR)

Agri-Biomass Biochar & Enriched Bio-Fertilizer — Green-Category Unit (starting 0.5 t/day, ramping to 2 t/day)

Promoter: Arun T. (sole proprietary concern — no partnership) **Location:** Eruthenpathy, Chittur Taluk, Palakkad District, Kerala (Kerala–Tamil Nadu border) **Feedstock origin:** Own coconut-farm residues + open-market agri/wood biomass procured from multiple local sources (no single-supplier or partnership dependency) **Structure:** A **single registered unit** that **starts small (~0.5 t/day) and ramps in place to ~2 t/day**; **non-core activities are distributed across the promoter's family land** (own plot, own farm and the promoter's mother's adjoining land), keeping the manufacturing core compact and in the **Green** pollution category. **Prepared for:** Submission to District Industries Centre (DIC), Palakkad and lending bank (AIF / term loan) **Document status:** Planning-grade DPR — figures are indicative estimates to be firmed up with (a) a feedstock proximate-analysis lab report, (b) formal machinery quotations, and (c) a written KSPCB categorisation confirmation, before sanction.

1. Executive summary

This project establishes an **agri-biomass biochar and enriched (microbe-inoculated) organic bio-fertilizer** business as a **sole proprietary concern of Arun T.**, as a **single unit that starts small and grows in place** on the promoter's family land at Eruthenpathy, Chittur, Palakkad.

- **Constitution:** sole proprietorship — **Arun T., proprietor**. No partnership, no partner-supplied feedstock.
- **One unit, phased in place (not split):** start at **~0.5 t/day biochar (500 kg/day)** and **ramp to ~1 t/day and then ~2 t/day** by adding modular reactor capacity within the **same registered premises** — no separate/duplicate units.
- **Feedstock: multi-source, own-farm + open-market agri/wood biomass** — coconut fronds, petioles, husk, shells and spathe from the promoter's own farm; coir pith and shells from local dehuskers/coir units; and open-market residues (sawdust/wood chips, cashew shell, areca husk, paddy husk, groundnut shell, prunings). Not limited to tender-coconut husk and not dependent on any one supplier.
- **Family-land layout:** the **manufacturing core** (pyrolysis + emission control + product store) stays on the **own 38-cent plot**; **non-core activities** (feedstock aggregation, sun-drying, chipping, bulk storage, enrichment/curing) are moved to the **promoter's own farm (~200 m)** and the **promoter's mother's adjoining land**, keeping the core small and lightly loaded.
- **Pollution category — Green (target).** Designed as *general biomass biochar (soil-amendment) production*, which the KSPCB revised categorisation list (as on 29.12.2025) places in the **Green** category — distinct from "charcoal from coconut shell by pyrolysis" (Orange). See Section 7.
- **Lenient consent:** starting small (**Phase-1 fixed capital ≤ ₹50 lakh**) brings the unit within KSPCB's **self-certification** thresholds for CTE/CTO (circular dated 16.09.2025) and the **micro-enterprise** band.
- **Institutional tailwind:** the project plugs into the **Kerala Government biochar expert committee** (constituted on a proposal of the **Australia India Business Council, AIBC**) and takes technical help from **CPCRI** and **KAU**, using the AIBC channel to explore **export to Australia and New Zealand**. See Section 16.
- **Phase-1 project cost:** **~₹52 lakh** (₹48 L fixed capital + ₹4 L working-capital margin); land is family-owned (in-kind, excluded from the ₹50 L capital test).
- **Financing:** promoter equity + **AIF term loan (3% interest subvention, 7 yrs)** + **CGTMSE** collateral-free cover + **Kerala ESS capital subsidy**.
- **Revenue stack:** enriched biochar sales (domestic + export) + **biochar carbon-removal credits** via an aggregator (Puro.earth CORCs).
- **Indicative payback:** ~2.5–3.5 years (Phase-1 physical-sales base case); under ~2 years if carbon-credit revenue is realised.

2. Promoter

Promoter / constitution: **Arun T. — sole proprietary concern** (proprietorship). This is **not a partnership**; there is no associate partner and no partner-supplied raw material.

- Family land in Eruthenpathy, Chittur Taluk, Palakkad: the promoter's **own 38-cent plot** (manufacturing core) and **own coconut farm (~200 m)**, plus the promoter's **mother's adjoining land** available for non-core use — together they comfortably host the unit and its feedstock/curing yards.
- Contact: 4/296, Mangapallam South, RVP Pudur Post, Palakkad District, Kerala – 678555 · 9901955667 · er.arunt@gmail.com.
- Captive demand: the promoter's own coconut farm consumes and demonstrates the enriched product, de-risking market entry.

3. Project concept & rationale

Agri-biomass residues (coconut fronds/husk/shell, coir pith, wood/sawdust and other crop residues) are abundant, low-value and often burnt in the open. Converting them to biochar via controlled, low-emission pyrolysis:

1. Diverts local biomass waste from open burning into a productive use.
2. Produces a stable, carbon-rich **soil amendment**; enriched with organics + microbes it becomes a branded slow-release **bio-fertilizer** (a soil input — **not** fuel charcoal and **not** activated carbon).
3. Generates durable **carbon-removal credits** (1 t biochar $\approx$ $\sim$ 2.2 t CO₂ sequestered for centuries).
4. Qualifies for strong central + state subsidy support as an agro-based / bio-fertilizer **micro** MSME.
5. Is structured to be **Green-category and (at the small start) self-certifiable**, so the unit can be established quickly and cheaply and then grown in place (Section 7).

4. Market analysis

Product (physical biochar), indicative prices:

Grade	₹/kg	25 kg bag
Raw bulk (agri)	15–40	₹375–1,000
Enriched / inoculated	50–120	₹1,250–3,000
Premium certified (EBC/organic)	up to 150	up to ₹3,750

Carbon credits: biochar CORCs trade ~US\$125–145 (Nasdaq CORCCHAR index); Indian industrial biochar clears ~€105–150/t CO₂. Biochar delivered ~43% of verified durable CDR volume; >90% of near-term supply is already contracted by buyers (Microsoft, Google, JPMorgan, Swiss Re) — a **supply-constrained, seller-favourable market**.

Export angle (Australia/NZ): Australia and New Zealand are active biochar markets — soil amendment, **livestock feed additive** (gut health, methane reduction), and carbon removal — coordinated by the **ANZ Biochar Industry Group (ANZBIG)**. This export channel is pursued through the AIBC/State-committee linkage (Section 16).

Demand strategy: anchor demand on (a) own-farm consumption, (b) enriched/branded regional sales, (c) export of physical product and the carbon credit. Sell *enriched* product (margin) rather than raw char (commodity).

5. Feedstock plan (multi-source, no partnership)

The unit is **not tied to a single feedstock or supplier**. It runs on a mix of own-farm and open-market biomass, giving supply security and reinforcing the *general biomass* (Green) classification.

Source	Material	Basis
Own coconut farm (200 m)	Fronds, petioles, husk, shells, spathe	Free / captive
Local dehuskers & coir units	Coir pith, coconut shells	Open-market purchase
Local sawmills / carpentry	Sawdust, wood chips, offcuts	Open-market purchase
Local farms / traders	Cashew shell, areca husk, paddy husk, groundnut shell, prunings, seasonal residue	Open-market purchase / collection

Supply security: execute **supply MoUs / purchase arrangements with 3–4 local sources** (not one partner) plus own-farm residue. Feedstock is documented **residual/waste biomass** for carbon additionality.

6. Technical design & capacity (single unit, phased in place)

6.1 Ramp path (one unit, capacity added in place — not split)

Phase	Biochar	Feed (dry biomass)	Annual biochar	CO ₂ removal
Phase-1 (start)	~0.5 t/day (500 kg/day)	~2 t/day	~150 t/yr	~330 t CO ₂ /yr
Phase-2	~1.0 t/day	~4 t/day	~300 t/yr	~660 t CO ₂ /yr
Phase-3 (endgame)	~2.0 t/day	~8 t/day	~600 t/yr	~1,320 t CO ₂ /yr

Capacity is grown by **adding modular reactor capacity within the same registered premises** and up-rating the shared gas train — **the unit is never split** into separate registrations.

6.2 Expandable-design principle ("start small, grow in place")

- **Reactor:** buy a small **~0.5 t/day** modular reactor now; add parallel/again-modular capacity for Phase-2/3 on **pre-laid foundations and flanged tie-ins**.
- **Shared infrastructure** (emission train, ID fan, stack, electrical, feed headers) is **installed for the near-term step and up-rated as the unit grows**, avoiding over-capitalising at the small start.
- **Consent** is taken in the Green category and **amended upward** as capacity ramps (Green throughout — no public hearing/EC).

6.3 Emission control & Green-category design

Closed, **continuous/semi-continuous** pyrolysis with a **syngas afterburner/flare** (self-fuelling the process and the dryer) → **cyclone/dust collector** → **alkaline scrubber with mist eliminator** → **stack** meeting KSPCB norms; **real-time temperature monitoring** and emergency shutdown; **Zero-Liquid-Discharge** (tar condensate and scrubber blowdown captured and reused — no effluent to land or water). **No open burning** at any stage. The design is built to the **KSPCB "Environmental Guidelines for Charcoal Manufacturing Units" dated 02.12.2025** (emission control, stack height, ash/dust housekeeping, heat reutilisation, responsible feedstock sourcing).

7. Pollution-control category & consent strategy

Objective: establish the unit in the Green category and (at the small start) via self-certification — the lenient regime — legitimately, on the merits of the process and the small scale.

7.1 Why Green applies

The KSPCB revised categorisation list **as on 29.12.2025** classifies *general charcoal / char (biochar) production* under **Green**, distinct from the *"Charcoal Production from Coconut Shell by Pyrolysis"* **Orange** entry (Notification No. KSPCB/159/2022-SEE-3 dated 10.08.2024). Our unit fits the **general / Green** description because:

- **Feedstock is mixed agri-biomass**, not a coconut-shell-only charge.
- **Product is biochar for soil**, not fuel charcoal and not activated carbon.
- **No pyrolysis-oil or carbon-black recovery** and **no chemical/steam activation**.
- **Low pollution potential** on the CPCB Pollution-Index basis — closed process with afterburner, scrubber, controlled stack emissions and **Zero-Liquid-Discharge**.
- **Guideline-compliant** with the 02.12.2025 charcoal-manufacturing guidelines.

7.2 Small start → self-certification

Category is set by the *activity*; the *scale* governs the **consent process**. Keeping **Phase-1 fixed capital ≤ ₹50 lakh** brings it within KSPCB's **self-certification** thresholds for CTE/CTO (circular dated 16.09.2025) and the **micro-enterprise** band. **Family-owned land is in-kind and excluded**, so it does not consume the ₹50 L headroom. Only the **manufacturing plot** (pyrolysis) needs CTE/CTO; the off-plot drying/storage/curing yards are **non-polluting ancillary** areas.

7.3 Consent path

- **Green + small Phase-1 → self-certification-based CTE and CTO** — fast, low-fee, no public hearing.
- **Ramp:** amend the Green consent upward for Phase-2/3 (still Green; no public hearing/EC).
- **No Environmental Clearance** (biochar from agri-biomass is not EIA-2006 scheduled; not Red).
- A **written KSPCB clarification** confirming the Green entry for this unit has been sought (separate category-clarification letter); sanction should follow that confirmation.

7.4 Honest caveats

- **Confirm the Green entry in writing** (KSPCB), and confirm whether the ₹50 L test is on *plant & machinery* or *total fixed capital*.
- **Family-land use:** obtain a **consent/lease/NOC** from the mother for the non-core yard on her land, and confirm that only the manufacturing plot triggers consent.

8. Machinery & suppliers (Phase-1 ~0.5 t/day, expandable)

Local (Coimbatore-Pollachi hub, 30-60 km — preferred for service/spares): coconut-shell/biomass carbonization plant manufacturers (e.g., Micro Fab Engineers, Essar Engineers, Veera Biopower and similar). Indicative small semi-auto units 0.5-1 t/day-class ≈ ₹6-25 L.

National (continuous, low-emission, certification-grade — better for EBC/carbon): - **Kerone Engineering (Mumbai)** — continuous multi-feedstock pyrolysis, waste-heat drying, cyclone + scrubber, SCADA. - **W2E Innovations (Ahmedabad/Kharagpur; Bengaluru branch)** — integrated biomass→biochar + wood-vinegar + fertilizer, "verification-

ready." - **Susstains Engineering (Karnataka)** — IIT-Madras-derived zero-smoke continuous process (~34% yield). - **Ylem Energy (New Delhi)** — smaller batch furnaces (lower cost; batch → less consistent).

Recommendation: buy a **compact continuous/semi-continuous ~0.5 t/day plant with integrated dryer + afterburner + scrubber** (not a bare batch kiln), with **pre-provisioned tie-ins for Phase-2/3 reactor modules** so the unit grows in place. Budget ~₹22 L for the reactor + gas train + dryer at Phase-1 (see Annexure D).

9. Land, layout & family-plot allocation

The business uses **three nearby family plots**, with only the **manufacturing core on the registered plot**:

Plot	Role	Activities
Own 38-cent plot (manufacturing unit)	Core (registered; needs CTE/CTO)	Pyrolysis reactor + emission control + product storage/bagging + office/lab/weighing
Own farm (~200 m)	Non-core	Feedstock aggregation, sun-drying yard , chipping/shredding, bulk biomass storage
Mother's adjoining land	Non-core	Enrichment / composting / curing yard , additional finished-goods & feedstock storage

Rationale: moving drying, storage, chipping and enrichment/curing **off the manufacturing plot** keeps the core compact and lightly loaded — good for fire-safety, for staying **Green/self-certifiable**, and for keeping bulk fuel away from the reactor. Only a 1–2 day day-bin sits on the core plot.

Indicative core-plot footprint (Phase-1): pyrolysis + emission control ~60 m²; day-bin ~30 m²; product store + bagging ~60 m²; office/lab/utilities ~60 m²; greenbelt/roads balance of the 38 cent.

Pre-checks: land-use where required; KSPCB setback from residences/water bodies/eco-sensitive zones; road access; 3-phase power; water; and a **written consent/lease/NOC** from the mother for use of her land.

10. Regulatory & statutory compliance

Approval	Authority	Notes
Udyam (MSME) registration	Online	Micro enterprise; gateway for schemes
Consent to Establish (CTE)	KSPCB	Green category; self-certification at the small Phase-1 start (≤ ₹50 L)
Consent to Operate (CTO)	KSPCB	Green; self-certification per applicable slab; amend upward on ramp
Panchayat D&O (Dangerous & Offensive) licence	Grama Panchayat	Manufacturing unit (core plot)
Building permit	Grama Panchayat	Shed/plant
Fire NOC	Fire & Rescue Dept.	Thermal process
GST registration		
Land-use / family-land NOC	Revenue Dept. / family	Core plot land-use; mother's land use consent

Operating without CTE/CTO is a criminal offence under EPA 1986. No open burning of biomass at any stage. Environmental Clearance is not required (non-EIA-scheduled, non-Red).

11. Cost of project (Phase-1, ~0.5 t/day)

#	Item	₹ Lakh
1	Core process shed + civil (compact, own plot)	9
2	Pyrolysis reactor ~0.5 t/day (modular, expandable)	16
3	Emission control — afterburner/flare + cyclone + scrubber + stack (Green-compliant)	6
4	Char cooling, sieving, enrichment mixer, bagging	5
5	Electrical, water, fire, weighing	4
6	Non-core on farm + mother's land: sun-drying floor, chipper/shredder, storage, curing yard	5
7	Lab / QC + office	1.5

#	Item	₹ Lakh
8	Preliminary & pre-operative (self-cert CTE/CTO, DPR, EBC/MRV)	1.5
	Fixed capital sub-total (Phase-1)	48
9	Margin money for working capital	4
	Total project cost (Phase-1)	~₹52 lakh

Family land is in-kind and **excluded** from the fixed-capital figure tested against the ₹50 L self-certification threshold.

Ramp add-ons (funded from cash flows / top-up AIF loan): Phase-2 (to 1 t/day) $\approx$ +₹28 L; Phase-3 (to 2 t/day) $\approx$ +₹55 L; cumulative endgame fixed capital $\approx$ ₹1.3 cr.

12. Means of finance & subsidy stack (Phase-1)

Source	₹ Lakh	%
Promoter equity (cash)	14	27%
Term loan — bank, under AIF (3% interest subvention, 7 yrs), CGTMSE collateral-free	34	65%
Working capital limit (bank cash credit)	4	8%
Total	52	100%

Kerala ESS capital subsidy (~₹20 L) — at the ~45% band on ₹48 L eligible fixed capital, within the ₹40 L cap — is received as **reimbursement after commissioning** and applied to **prepay the term loan**, cutting effective debt to ~₹14 L.

Subsidy stack logic

- **Anchor = Kerala ESS** (state capital subsidy). Target **45%** band: 25% base (young/women/SC-ST/NRK) + 10% priority sector (agro-based / bio-fertilizer) + 10% new technology. *Palakkad does not get the extra 10% district bonus.*
- **+ AIF** (central): 3% interest subvention + CGTMSE guarantee — different basis, **stacks legitimately with ESS**; confirm convergence in writing with DIC + bank.
- **+ CGTMSE:** collateral-free loan → the land is not mortgaged.
- **Do NOT use PMEGP** — its rules disqualify units that have availed other government subsidy, voiding ESS + AIF.
- **Effective term-loan interest:** 9% – 3% = ~6%.
- **Ramp capital** (Phase-2/3) is funded from cash flows + a top-up AIF loan, keeping Phase-1 the only external ask.

13. Working capital (indicative, annual, Phase-1)

Feedstock collection/purchase/transport, enrichment inputs (poultry manure, neem cake, rock phosphate, microbial consortium), packaging, wages, power, and ~1–2 months finished-goods holding. Phase-1 operating cost $\approx$ **₹30 L/yr**; scales with capacity on ramp.

14. Profitability & ROI (Phase-1 and endgame)

Common assumptions: 300 operating days; biochar 1 t $\approx$ 2.2 t CO₂; carbon net realization $\approx$ ₹10,000/t CO₂ (after aggregator/MRV); enriched blended price ₹35/kg. Effective interest 6%.

Scenario	Biochar	Revenue	Op. cost	Net profit (approx)	Payback
Phase-1 — base (physical only)	150 t	₹52.5 L	₹30 L	~₹12 L	~2.5–3.5 yr
Phase-1 — + carbon	150 t	₹52.5 L + ₹33 L = ₹85.5 L	₹32 L	~₹40 L	~1.2–1.5 yr
Endgame 2 t/day — base (physical only)	600 t	₹2.10 cr	₹1.05 cr	~₹80 L	—
Endgame 2 t/day — + carbon	600 t	₹2.4 cr + ₹1.32 cr = ₹3.72 cr	₹1.20 cr	~₹2.1 cr	<1 yr

Net capital at risk (Phase-1) $\approx$ ₹28 L after ESS $\sim$ ₹20 L. **ROI (Phase-1, base): $\sim$ 20%; with carbon: $\sim$ 85%.** Carbon revenue is upside — treat physical-only as the bankable base case; carbon requires certification + a signed aggregator offtake.

Drivers: small low-risk start, low-cost multi-source feedstock, ~6% subsidised interest, ESS, selling enriched (not raw) product, and growing capacity in place from cash flows.

15. Carbon-credit strategy & aggregator options

At ~330 t CO₂/yr (Phase-1) rising to ~1,320 t CO₂/yr (endgame), volumes are **too small to transact directly with corporate buyers** initially (~85% of CORCs are sold via brokers/aggregators). Route through an **aggregator/developer** who provides MRV, certification (Puro.earth CORC methodology) and buyer access.

Party	Role / fit
Varaha (Gurugram/New Delhi) — India's largest biochar CDR developer; Puro.earth + Verra; offtakes with Google, Microsoft, Kellanova	Best first contact. Its Industrial Partners Program lets facilities with feedstock + conversion capability use Varaha's digital MRV + credit commercialization and issue CORCs.
Equilibrium — industrial biochar operator	Developer/partner; potential offtake channel.
Eco Saarathi — cooperative aggregating agri residue → biochar	Fit for an aggregation structure.
Puro.earth	The registry/standard (CORCCHAR methodology); certification via a developer.
BioFlux (advisory)	Independent biochar-CDR advisory — feasibility, MRV & certification, buyer alignment.

Action: approach **Varaha first** for MRV + offtake; target a **Letter of Intent (LOI)** as volumes ramp. Document feedstock as **residual/waste biomass** for additionality.

16. Institutional collaboration & export — AIBC, the Kerala biochar expert committee, CPCRI/KAU, and Australia/NZ

Kerala's biochar push has a **direct Australia linkage** the project should leverage:

- **How it began:** the **Australia India Business Council (AIBC)** — which promotes Australia-India bilateral trade — sent a proposal (letter dated **15 July**) highlighting biochar's carbon-sequestration and trade potential. On that basis the **Kerala Government constituted a 7-member biochar expert committee, chaired by the Special Officer for Agriculture (WTO Cell)**, with experts from the **Kerala Agricultural University (KAU)** and the **Central Plantation Crops Research Institute (CPCRI)**, and officials of the Agriculture Development & Farmers Welfare Department (with inputs from Oil Palm India Ltd). Its brief: study **biochar and derivatives (charcoal briquettes, activated charcoal) from coconut biomass**, for **domestic use and export**, with **Australia and New Zealand** flagged as target markets (animal feed, sustainable farming).

How this project takes help from them:

1. **Plug into the State expert committee (WTO Cell).** Write to the Special Officer for Agriculture (WTO Cell) offering this unit as an **on-ground pilot / demonstration** feeding the committee's feasibility study. Benefit: State recognition, policy alignment (including the **Green-category** treatment), visibility for incentives, and a credible reference for buyers and lenders.
2. **Technical help from CPCRI (Kasaragod) and KAU (Vellayani).** CPCRI has pyrolysed **coconut husk, petiole and coir-pith into biochar** (characterisation and soil-amendment validation); KAU has field trials of **coconut-frond biochar + liming on Kerala laterite soils**. Seek their support for **feedstock-specific biochar recipes, product characterisation/QC, agronomic validation and farmer demonstrations** — this de-risks quality, speeds EBC/organic certification, and builds buyer confidence.
3. **Market promotion & export via the AIBC.** Engage the **AIBC** (the initiator of the State initiative) as a **trade facilitator** to access **Australia/NZ buyers** and position the product for **animal-feed-grade and soil-amendment export** — becoming an early industry participant/case study in the AIBC's biochar-trade channel.
4. **Standards & biosecurity for actual export.** Align product to the **ANZBIG (Australia-NZ Biochar Industry Group) Code of Practice** and EBC; and clear **Australian biosecurity import (BICON)** — fully carbonised biochar from plant material falls under the "Plant derived charcoal, wood pellets, briquettes and firewood" case and typically needs an **import permit** and compliance with biosecurity conditions before shipment.

Actions: (a) letter to the Special Officer (WTO Cell) to enrol as a pilot; (b) technical-support/MoU requests to CPCRI and KAU; (c) engage the AIBC for Australia/NZ market access; (d) begin ANZBIG/EBC alignment and a BICON import-permit enquiry for a first Australia/NZ consignment.

17. Value addition — enriched bio-fertilizer

Base biochar is K-rich; enrich to a balanced slow-release organic input: - **Nitrogen/phosphorus:** poultry manure, neem cake, bone meal / rock phosphate. - **Biology (branded premium):** microbial consortium — Azospirillum/Azotobacter (N-fix), PSB/phosphobacteria, potash mobilizers, Trichoderma, mycorrhiza (VAM). - **Base/biology co-compost:** vermicompost + cow dung; Kerala liquid organics (fish amino acid, panchagavya, jeevamrutham); seaweed extract for micronutrients. - **Methods:** nutrient impregnation/soaking then low-temp drying; or co-composting/vermicomposting with ~10% biochar (on the mother's-land curing yard); pelletize to cut leaching. - **Certifications for premium/export:** EBC, IBI, ANZBIG Code of Practice (for Australia), organic-input (Jaivik Bharat / NPOP for export).

18. Implementation schedule (sequence, not calendar)

1. Feedstock lab proximate analysis (moisture, fixed carbon, ash, K) — decides final Phase-1 sizing.
2. Udyam registration; DIC/ESS pre-consultation; feedstock MoUs (3–4 local sources); **mother's-land use NOC**.
3. **Written KSPCB categorisation confirmation** (Green); enrol with the **WTO-Cell expert committee** as a pilot.
4. **Self-certified CTE (Green)**; machinery order (~0.5 t/day reactor + gas train + dryer, with ramp tie-ins).
5. Compact civil on the core plot; non-core yards on farm + mother's land.
6. Panchayat D&O + building permit + fire NOC; **self-certified CTO**.
7. AIF term loan + CGTMSE; commissioning; ESS claim.
8. Product validation on own farm; **CPCRI/KAU** characterisation; EBC/organic (and ANZBIG) certification.
9. Engage carbon aggregator (Varaha) and the **AIBC** for Australia/NZ market access; BICON enquiry.
10. **Ramp in place** to 1 t/day then 2 t/day from cash flows / top-up loan.

19. Risks & mitigation

Risk	Mitigation
Pollution category	Design for Green on the merits; written KSPCB confirmation ; keep Phase-1 small (self-cert)
Family-land use	Written consent/lease/NOC from the mother; only the core plot needs consent
Feedstock moisture / drying energy	Sun-dry at farm/mother's land; waste-heat dryer; lab-test first
Feedstock security	Multi-source (own farm + 3–4 open-market suppliers) — no single-supplier dependency
Demand-side weakness (Indian biochar)	Own-farm use; sell enriched/branded; export via AIBC to Australia/NZ; carbon credit
Export/biosecurity (Australia)	Early BICON import-permit enquiry; ANZBIG/EBC certification
Carbon-credit dependency	Bank case = physical only; carbon = upside pending LOI
Product consistency for certification	Continuous reactor + fixed blend recipe; CPCRI/KAU validation

20. Assumptions & disclaimer

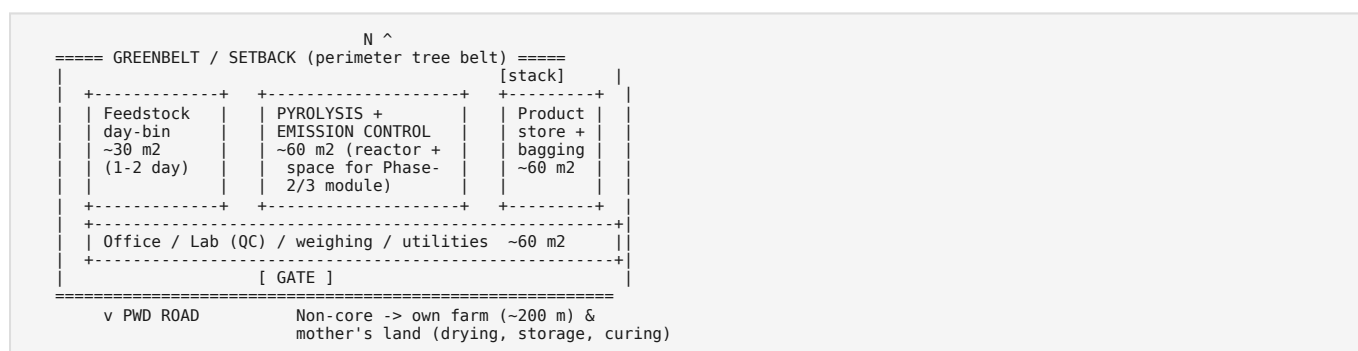
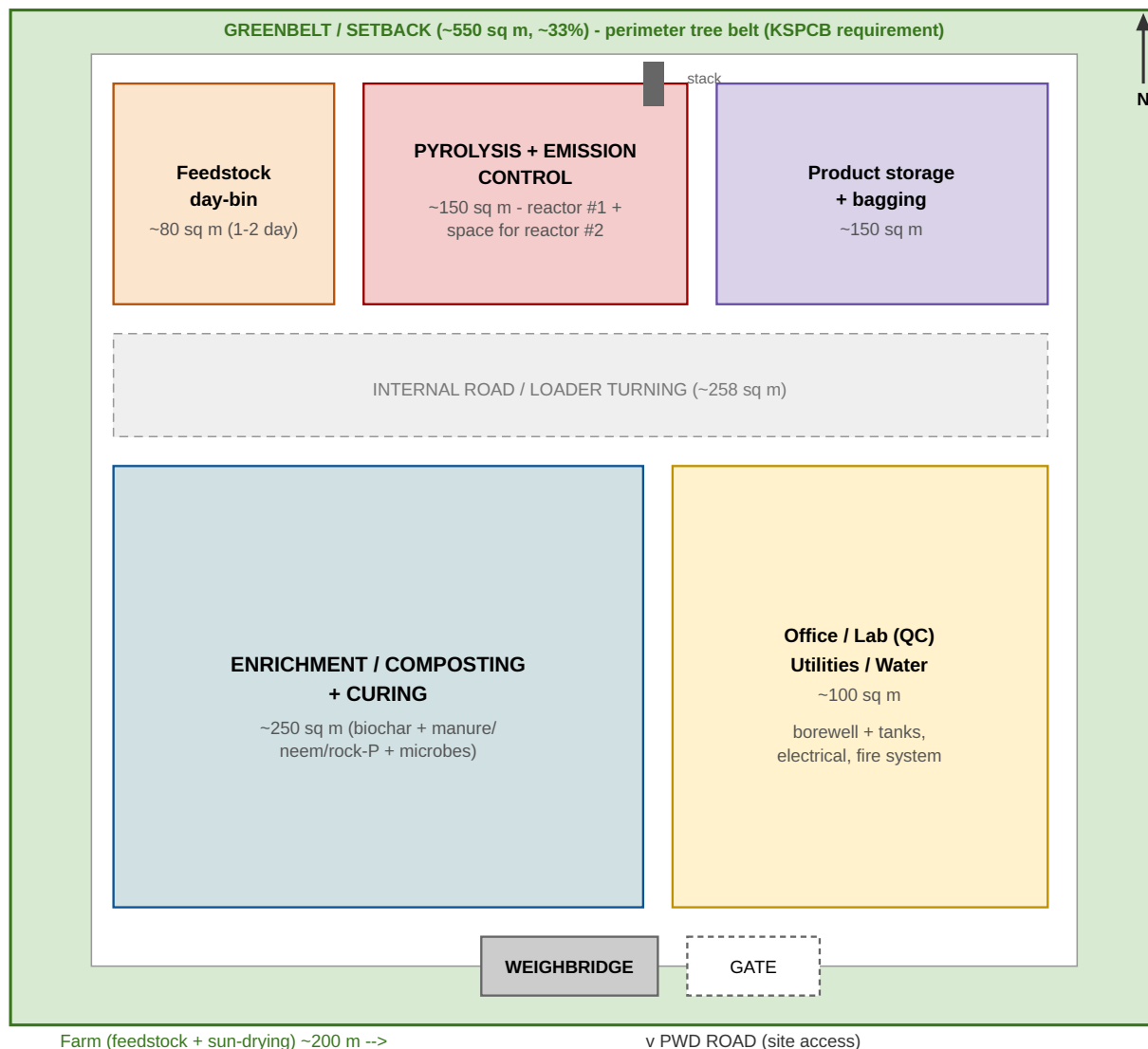
All costs, yields, prices and returns are **planning-grade indicative estimates** (mid-2026 market references) for scheme application and bank appraisal. They must be confirmed by: (a) a laboratory proximate analysis of the actual feedstock; (b) formal machinery quotations; (c) written confirmation of ESS/AIF eligibility and convergence from DIC Palakkad and the lending bank; (d) a **written KSPCB categorisation (Green) and self-certification confirmation**; (e) a **mother's-land use consent/NOC**; and (f) a carbon-credit LOI and BICON/ANZBIG export-compliance confirmation before export/carbon revenue is counted. Regulatory categories and subsidy rates are subject to the prevailing KSPCB, DIC and scheme guidelines at the time of application.

Annexure A — Core-plot layout (Phase-1 micro-footprint, expandable in place)

plot-layout.svg (open in any browser) shows the owned site. Text schematic of the **Phase-1 core plot** below:

Plot Layout - 38 cent (~40 m x 38.5 m ~ = 1,538 sq m)

Biochar and enriched bio-fertilizer unit - schematic (2 t/day-ready), not strictly to scale



Notes: the core plot carries only the reactor + emission train + product store + office; a vacant bay with pre-laid foundation/tie-ins takes the **Phase-2/3 reactor module** (grow in place). Bulk biomass drying/storage and enrichment/curing are on the **own farm and the mother's adjoining land**, so only a 1-2 day day-bin sits on the core plot.

Annexure B — Term-loan amortisation (Phase-1)

Assumptions: term loan ₹34 L; effective rate 6% p.a. (9% – 3% AIF subvention); tenure 7 years; 6-month construction moratorium (interest served, no principal in Year 1); Kerala ESS ~₹20 L received ~Month 12 and applied as principal prepayment, leaving ~₹14 L to amortise over Years 2-7 (EAI ≈ ₹2.85 L/yr, ≈ ₹0.24 L/month).

Year	Opening (₹L)	Interest @6% (₹L)	Principal repaid (₹L)	ESS prepayment (₹L)	Closing (₹L)
1	34.00	2.04	0.00 (moratorium)	20.00	14.00
2	14.00	0.84	2.01	-	11.99
3	11.99	0.72	2.13	-	9.86

Year	Opening (₹L)	Interest @6% (₹L)	Principal repaid (₹L)	ESS prepayment (₹L)	Closing (₹L)
4	9.86	0.59	2.26	–	7.60
5	7.60	0.46	2.39	–	5.21
6	5.21	0.31	2.54	–	2.67
7	2.67	0.16	2.69	–	~0.00
Total		~5.1	~14.0	20.0	

AIF benefit: the 3% subvention saves ~₹2.5 L of interest over the loan life; the loan is **collateral-free** under CGTMSE — the land is not mortgaged.

Annexure C — Month-by-month cash flow, Year 1 (Phase-1 ramp)

Assumptions (physical-sales base case): commissioning at **Month 5**; ramp 30% → 100% by Month 10. Full-capacity monthly revenue **₹4.38 L** (₹52.5 L/yr); operating cost ≈ **₹1.2 L fixed + ₹1.3 L variable at 100%**; interest served during moratorium ≈ **₹0.17 L/month**. Capex is funded by equity + term loan (not shown in operating flow). ESS ₹20 L inflow (~Month 12) is a financing item applied to loan prepayment (Annexure B).

Month	Phase	Capacity	Revenue (₹L)	Op. cost (₹L)	EBITDA (₹L)	Interest (₹L)	Net op. cash (₹L)
1–4	Construction / pre-op	–	0.00	trial only	–	0.17/mo	interest served from loan
5	Commissioning	30%	1.31	1.59	–0.28	0.17	–0.45
6	Ramp	45%	1.97	1.79	0.18	0.17	0.01
7	Ramp	60%	2.63	1.98	0.65	0.17	0.48
8	Ramp	75%	3.28	2.18	1.10	0.17	0.93
9	Ramp	90%	3.94	2.38	1.56	0.17	1.39
10–12	Full	100%	4.38/mo	2.50/mo	1.88/mo	0.17	1.71/mo

Steady state (Year 2+, Phase-1): monthly revenue ₹4.38 L – opex ₹2.50 L = **EBITDA ₹1.88 L**; less EMI ₹0.24 L → **net cash ~₹1.64 L/month (~₹19.6 L/yr)** before tax and depreciation.

Debt-service coverage (Year 2): EBITDA ≈ ₹22.5 L ÷ debt service ≈ ₹2.85 L → **DSCR ≈ 7.9** (very comfortably bankable).

Upside: adding realized carbon revenue (~₹33 L/yr at Phase-1) lifts EBITDA sharply and shortens payback to well under ~1.5 years — kept out of the base case above pending a signed aggregator LOI.

Annexure D — Machinery quotation-comparison template (fillable, Phase-1 ~0.5 t/day)

How to use: send this to at least **two local vendors (Coimbatore/Pollachi hub)** and **one national continuous-technology vendor**. Ask each to quote a complete **~0.5 t/day plant** with **pre-provisioned tie-ins for Phase-2/3 modules**, state guaranteed emission values and yield, then carry the total into **Section 11 — Cost of project**.

Table D-1 — Scope & required specification (issue to vendors)

#	Equipment	Sized for	Required specification / notes	→ Cost item
1	Pyrolysis / carbonization reactor	~0.5 t/day (~100 kg/hr feed), expandable	Continuous or semi-continuous; multi-feedstock (husk, frond, shell, coir, sawdust); yield ≥25–30% dry basis; pre-laid tie-ins for Phase-2/3 module	#2
2	Feed system (hopper, screw/belt, metering)	0.5 t/day	Variable-speed; suited to shredded mixed biomass	#2
3	Waste-heat feed dryer	0.5 t/day	Uses pyrolysis/syngas waste heat; inlet moisture up to ~25%	#2
4	Emission control — afterburner/flare + cyclone + alkaline scrubber + mist eliminator + ID fan + stack	0.5 t/day (up-ratable)	Must meet KSPCB Green / 02.12.2025 guideline stack norms; vendor to state guaranteed emission values & stack height	#3
5	Char cooling / quench + discharge	0.5 t/day	Sealed cooling; moisture/temperature controlled	#4
6	Sieving / screening	0.5 t/day	Grade separation (powder/granular)	#4
7	Enrichment mixer / blender	0.5 t/day	For char + manure/neem/rock-P + microbial slurry	#4

#	Equipment	Sized for	Required specification / notes	→ Cost item
8	Weighing + bagging	0.5 t/day	25 kg bags + FIBC jumbo option	#4
9	Control panel / instrumentation	0.5 t/day	Temp/pressure monitoring, interlocks, e-stop	—
10	Erection, commissioning & operator training	—	Include site supervision & trial run	#2-#4
11	Freight, insurance, unloading	—	To site, Eruthenpathy	#2-#4
12	Taxes (GST)	—	State applicable rate separately	—

Table D-2 — Price comparison (fill from vendor quotes, ₹)

#	Equipment (as above)	Vendor A: _____	Vendor B: _____	Vendor C: _____
1	Pyrolysis reactor (0.5 t/day, expandable)	₹ _____	₹ _____	₹ _____
2	Feed system	₹ _____	₹ _____	₹ _____
3	Waste-heat dryer	₹ _____	₹ _____	₹ _____
4	Emission control train	₹ _____	₹ _____	₹ _____
5	Char cooling / quench	₹ _____	₹ _____	₹ _____
6	Sieving / screening	₹ _____	₹ _____	₹ _____
7	Enrichment mixer	₹ _____	₹ _____	₹ _____
8	Weighing + bagging	₹ _____	₹ _____	₹ _____
9	Control panel	₹ _____	₹ _____	₹ _____
10	Erection, commissioning, training	₹ _____	₹ _____	₹ _____
11	Freight, insurance	₹ _____	₹ _____	₹ _____
	Sub-total (ex-GST)	₹ _____	₹ _____	₹ _____
12	GST	₹ _____	₹ _____	₹ _____
	TOTAL (incl. GST)	₹ _____	₹ _____	₹ _____
	<i>Target ≈ ₹22 L for reactor+dryer+emission (items #2-#3 of Section 11)</i>			

Table D-3 — Commercial-terms comparison (fill from vendor quotes)

Parameter	Vendor A	Vendor B	Vendor C
Make / model offered			
Rated capacity (kg/hr feed; t/day biochar)			
Guaranteed biochar yield (% dry basis)			
Expandability to 1-2 t/day (tie-in provisions)			
Material of construction (reactor)			
Connected electrical load (kW)			
Guaranteed stack emissions vs KSPCB Green/guideline norms			
Feedstock types accepted (husk/frond/shell/coir/sawdust)			
Delivery period (weeks from PO)			
Warranty (months)			
After-sales / spares — distance from site			
Payment terms			
Training & commissioning included?			
References / plants in operation (contactable)			

Evaluation guidance: do not select on price alone — weight **emission-compliance (for Green CTO), guaranteed yield, feedstock flexibility, expandability in place, and after-sales proximity** (favouring the Coimbatore/Pollachi hub). A clean,

low-emission continuous unit both satisfies the Green/guideline norms and produces consistent, certifiable biochar for domestic sale and Australia/NZ export.